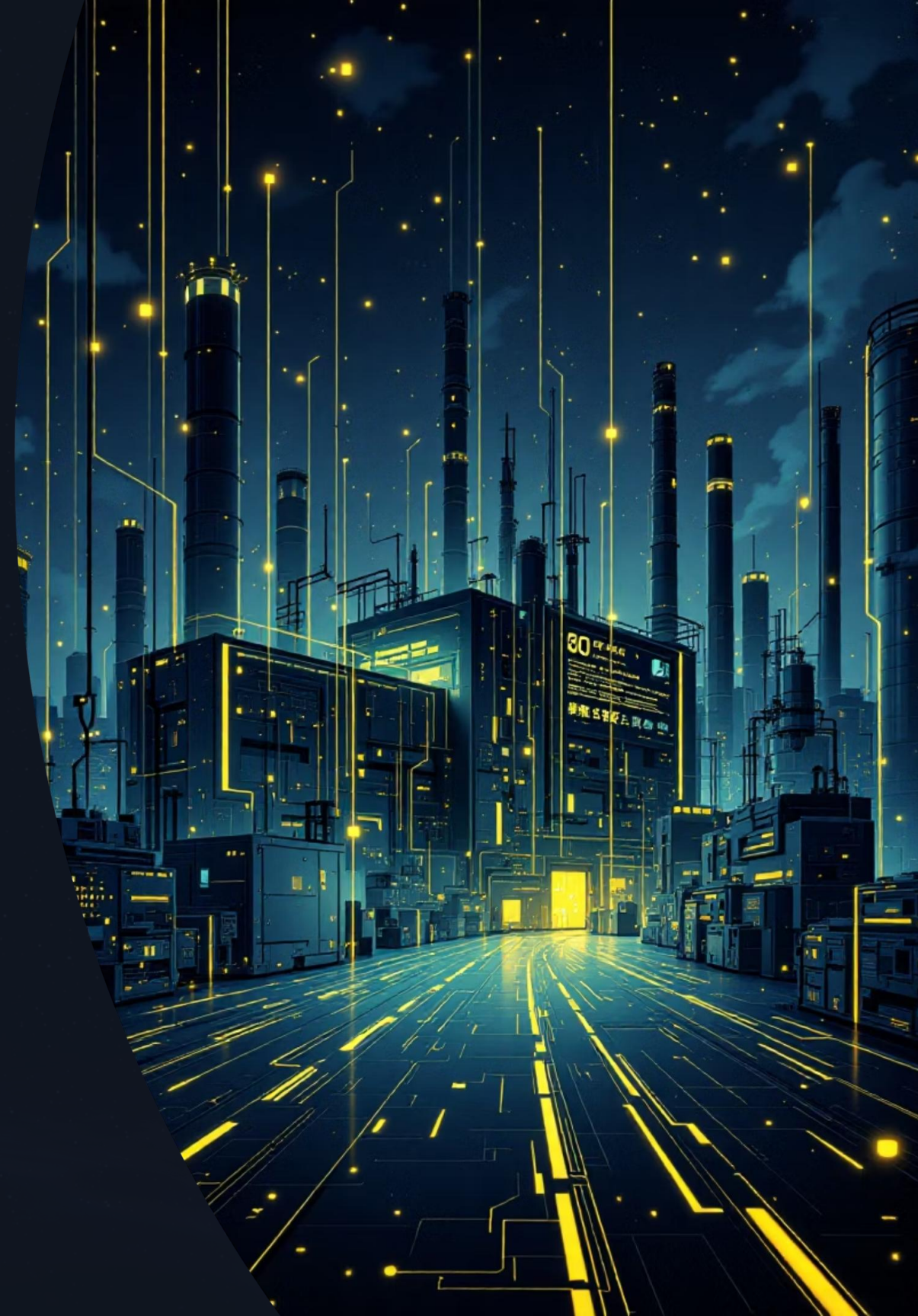


SmartBuild Optimization: Predictive Quality Assurance

Presented by Shahmir Zaman, Ishan , Muhammad Ahmad





Executive Summary: Maximizing Value Through Prediction

Current Cost of Defect

Each detected manufacturing defect incurs a cost of **150 EUR**, accounting for both machine downtime and material waste.

Cost of Prevention

By predicting and discarding faulty raw material at the input stage, the cost drops to a mere **10 EUR** per potential defect.

Business Value

Our predictive models enable SmartBuild to save **140 EUR** for every error detected and prevented at the earliest stage.

Data & Methodology: The Foundation of Insight

Our approach leverages SmartBuild's existing data to build robust predictive models.



Input Data Collection

Raw data from **Machine Settings Logs** and **Production Logs** provides a comprehensive view of manufacturing processes.



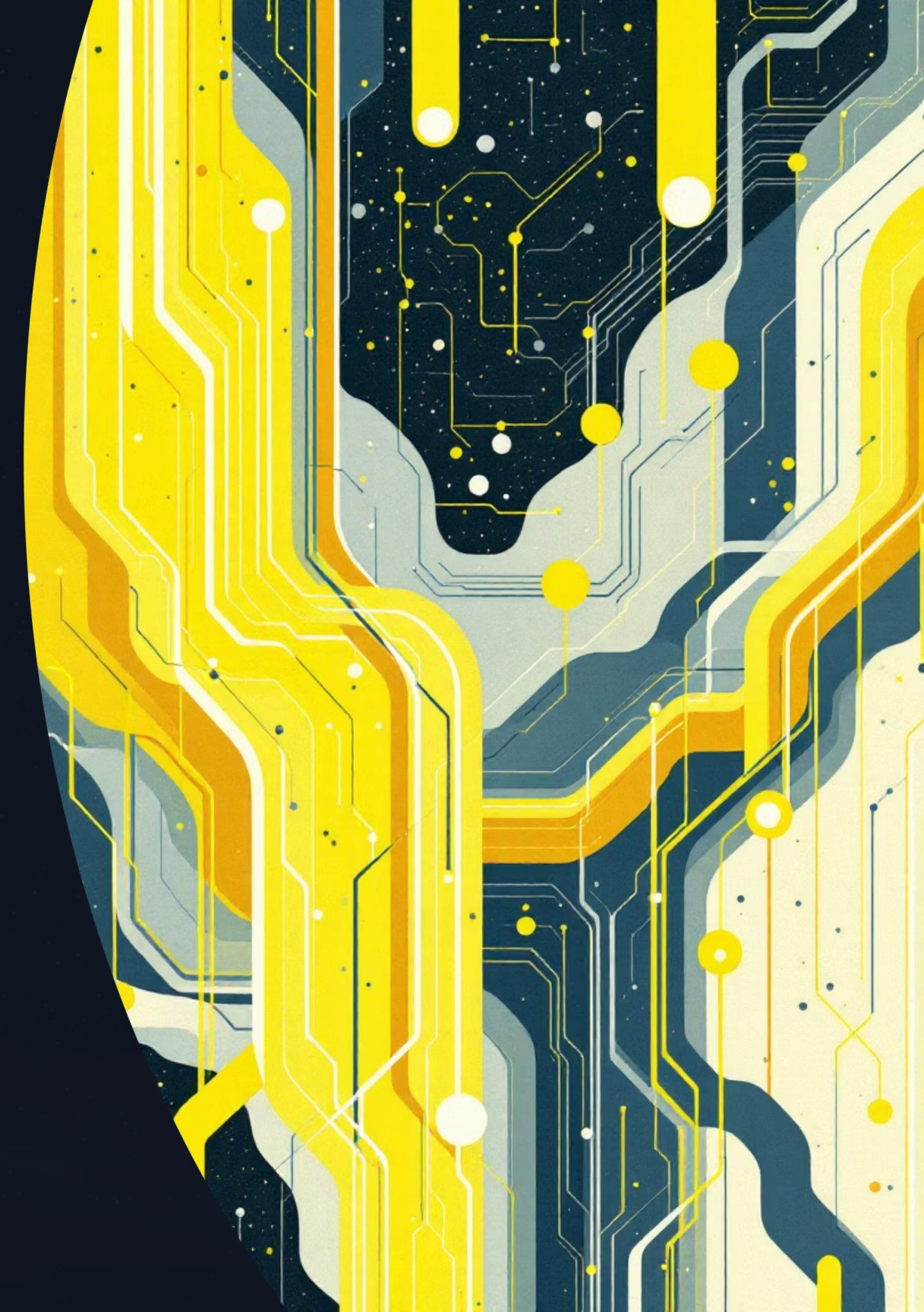
Data Processing & Cleaning

Utilizing Python, raw data undergoes rigorous cleaning, transformation, and feature engineering to prepare it for model training.



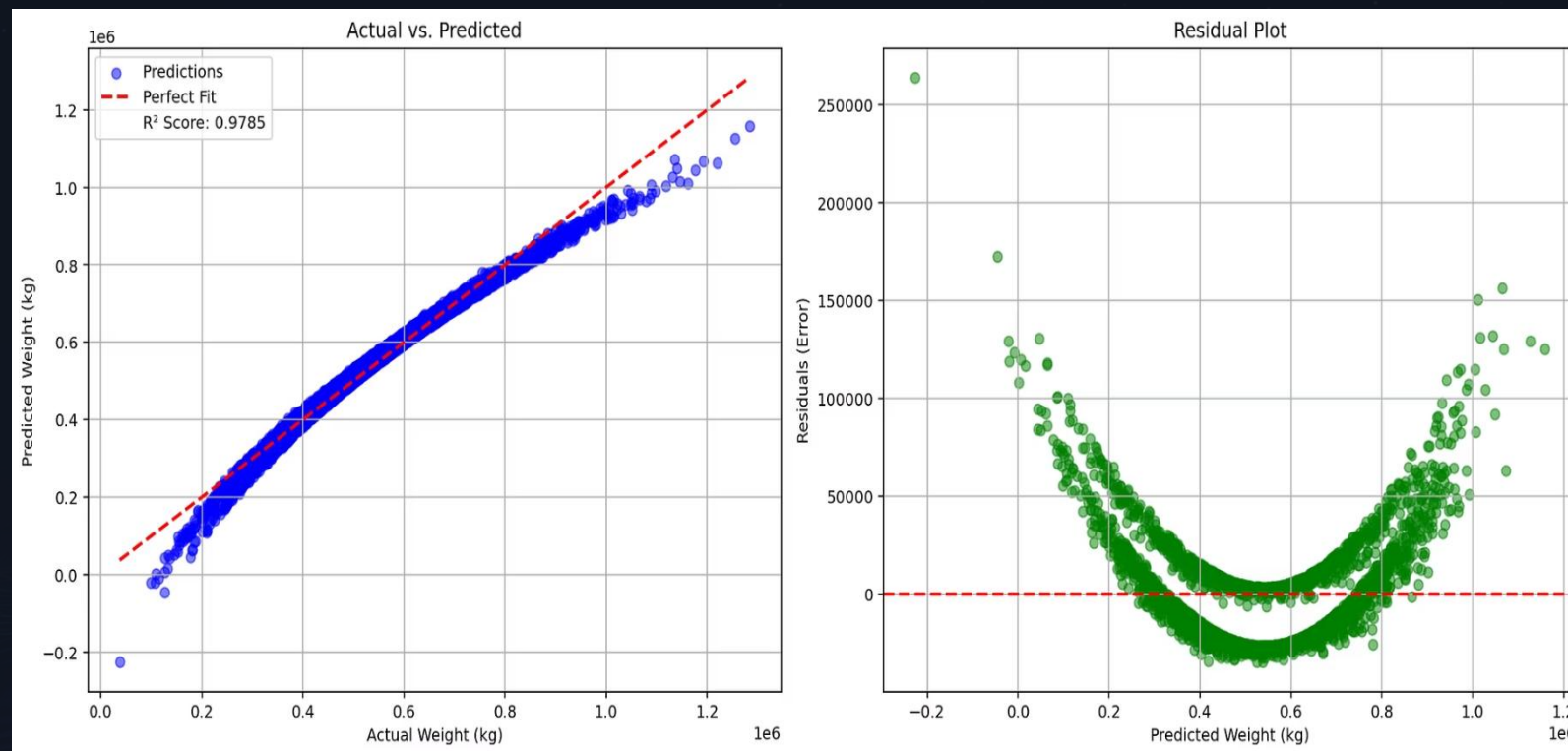
Model Development

We developed two key models: **Weight Prediction (Regression)** to forecast product weight and **Error Prediction (Classification)** to identify defect types.



Weight Prediction: Initial Approach with Linear Regression

Our first attempt at predicting product weight utilized a standard Linear Regression model.



Model Performance

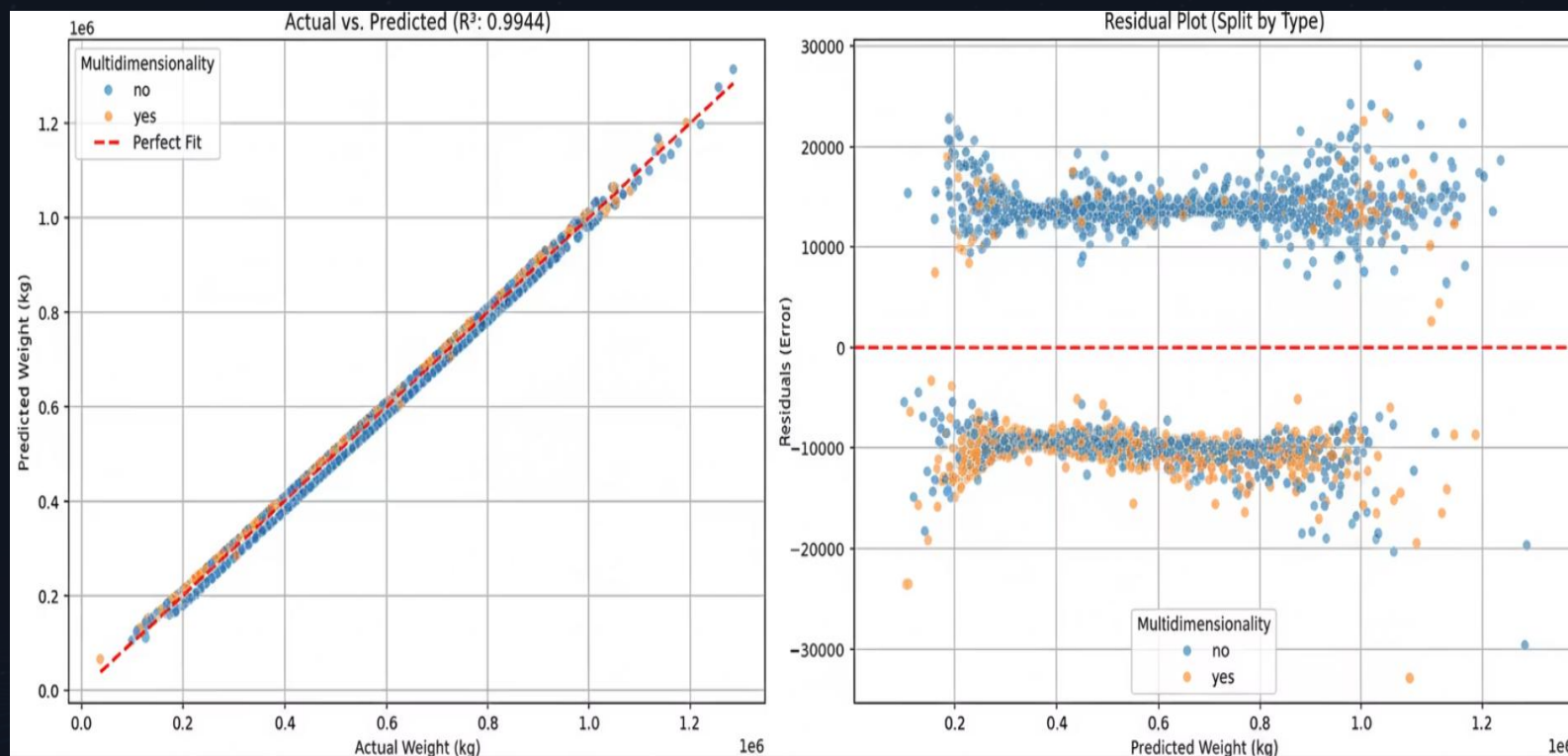
The Linear Regression model achieved a strong R^2 of 0.98, indicating a good overall fit.

Key Observation

Despite the high R^2 , analysis of the residuals revealed a distinct "U-Shape", indicating a systematic error not captured by the linear model.

This systematic pattern suggested that a more complex relationship was at play, requiring a different modeling approach.

Task 1: Uncovering Hidden Interactions



Model

Polynomial Regression ($R^2 > 0.99$)

Residual Analysis

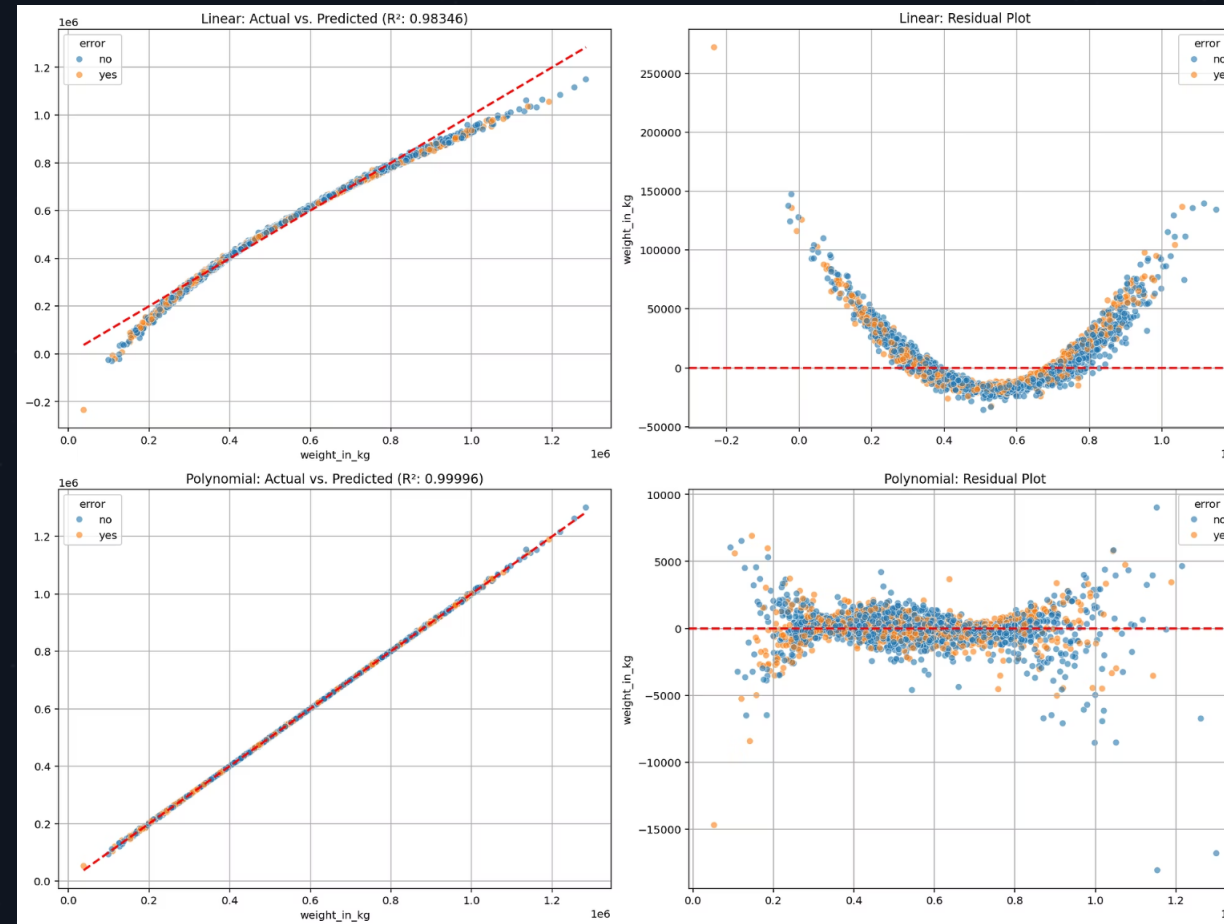
The "U-shape" is eliminated, but a distinct "Split" pattern remains.

Insight

This split is not random. It represents specific machine states driven by a combination of FluxCompensation, IonizationClass, and Error status.

Eliminating Systematic Bias: Linear vs. Polynomial

Visualizing the distinct improvements achieved by moving from a linear to a polynomial regression model for weight prediction.



Linear Model: Unacceptable Bias

- Displays a clear "U-Shape" in residuals.
- **Risk:** The model consistently underestimates weight at the extremes (low/high inputs), leading to systematic errors.

Polynomial Model: Enhanced Accuracy

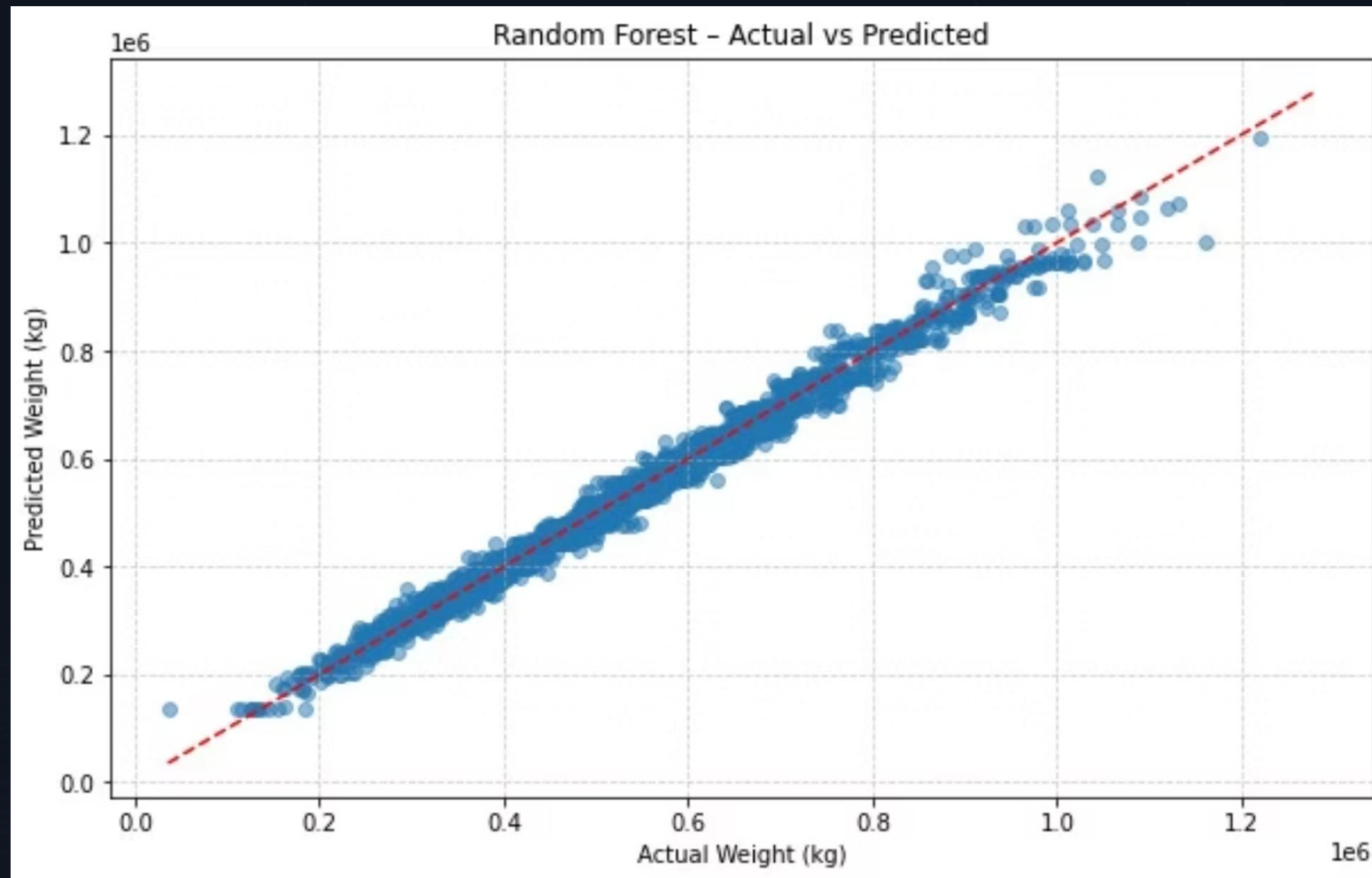
- The "U-Shape" is completely flattened, indicating the model now captures the underlying non-linear relationships.
- **Result:** Accuracy increased significantly, achieving an R^2 greater than 0.99, demonstrating robust predictive capability.

Residual Split: Operational Insights

- The distinct "Residual Split" in the polynomial model visualizes the machine's different operating modes (Normal vs. Error states).
- This insight confirms our model successfully captures these critical operational nuances, paving the way for more targeted interventions.

Weight Prediction: Random Forest Verification

To ensure the robustness of our chosen weight prediction model, we benchmarked it against a Random Forest algorithm.



Comparative Analysis

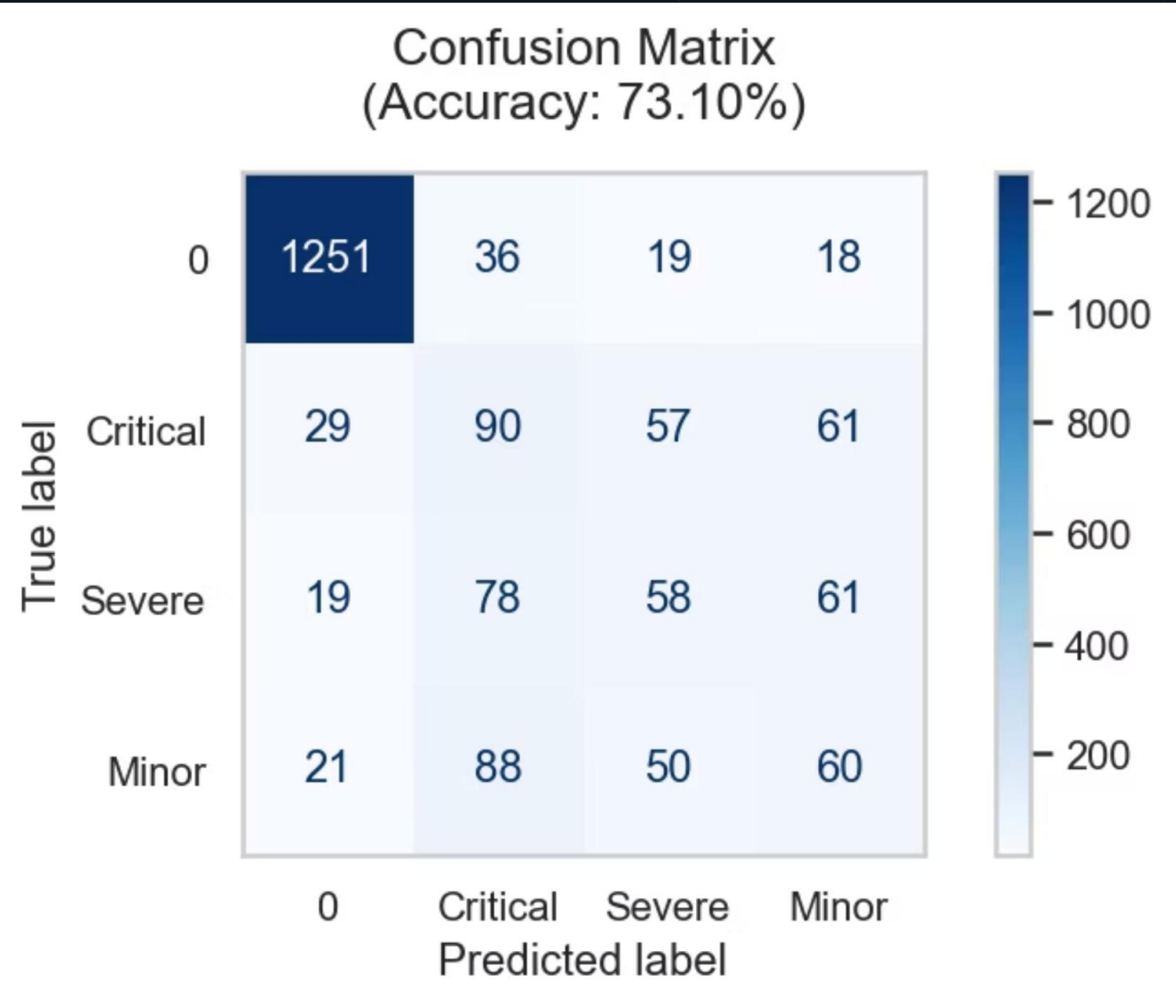
The Random Forest model also performed well, confirming the feasibility of accurate weight prediction.

Model Selection

For this specific regression task, Polynomial Regression proved **superior** in terms of interpretability and predictive power for the identified data patterns.

While Random Forest is a powerful general-purpose model, Polynomial Regression offered the optimal solution for this particular scenario.

Error Prediction: Baseline Model Performance



Model: Random Forest Classifier

Our initial error prediction utilized a Random Forest Classifier to identify defective products.

Overall Accuracy: 73.10%

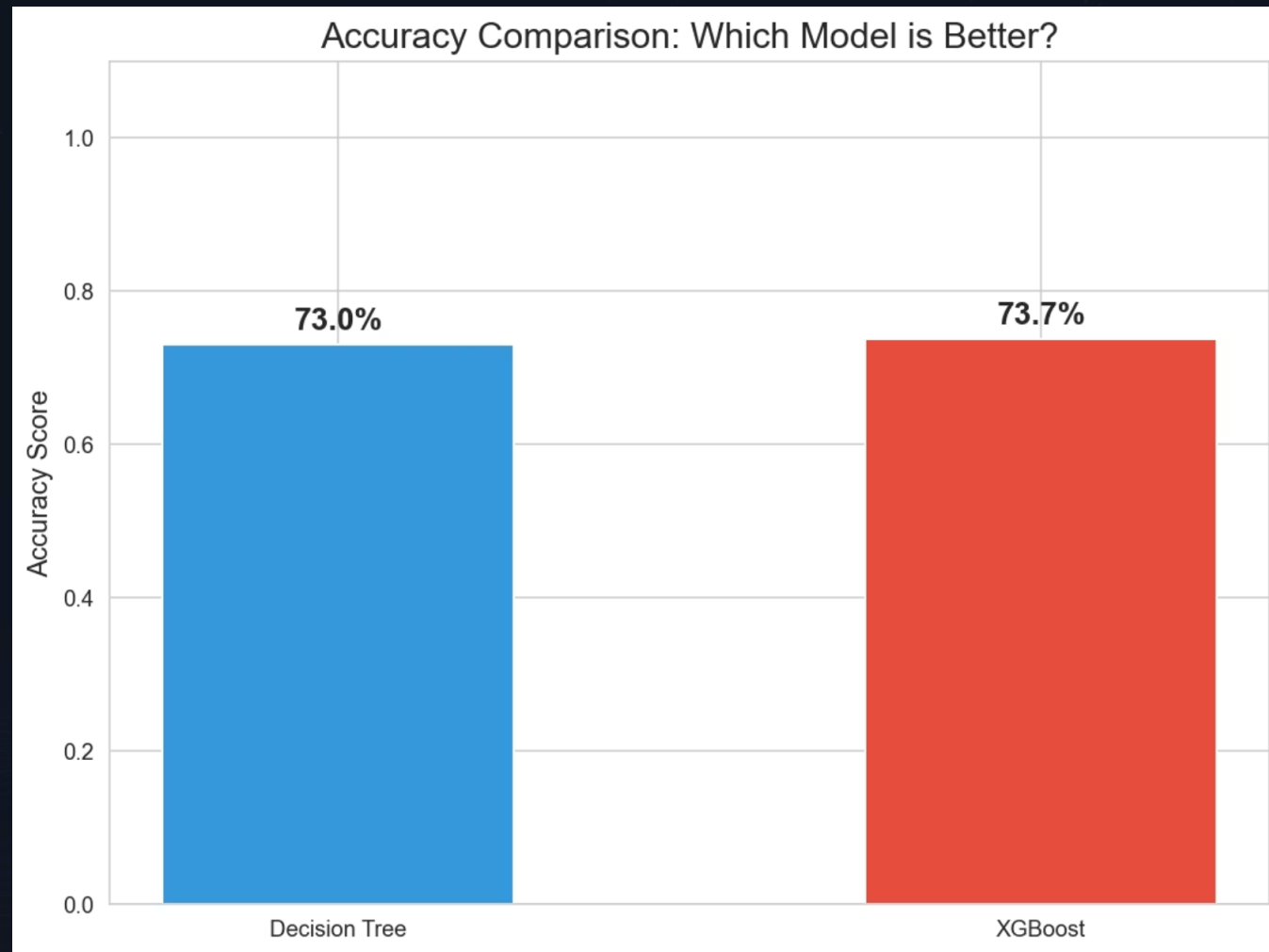
The model demonstrated a decent overall performance in classifying product quality.

Limitations: Error Type Differentiation

A primary challenge was its struggle to accurately differentiate between specific error types: **Critical**, **Severe**, and **Minor** defects.

Classification Model Comparison

Our objective is to accurately predict the `error_type` at an early stage, minimizing waste and costs.



Decision Tree Accuracy

A baseline Decision Tree Classifier achieved an accuracy of 73.0%.



XGBoost Performance

The XGBoost Classifier significantly improved accuracy to 73.7%, demonstrating its superior predictive capabilities for defect classification.



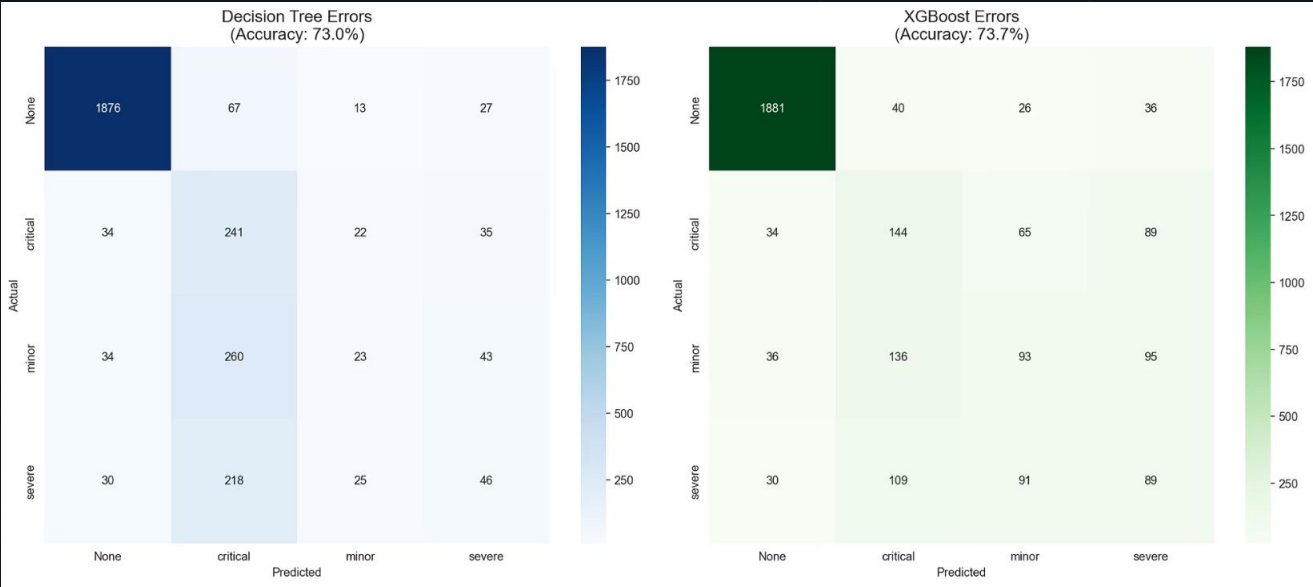
Selected Model

Given its higher accuracy and robustness, XGBoost has been selected as the primary model for error type prediction.

This 0.7% improvement translates to substantial cost savings when applied across high-volume manufacturing.

Financial Impact: Maximizing Net Savings

The implementation of our XGBoost classification model yields substantial financial benefits for SmartBuild.



Status Quo Cost (No Model) 151650

Cost with XGBoost Model 25130

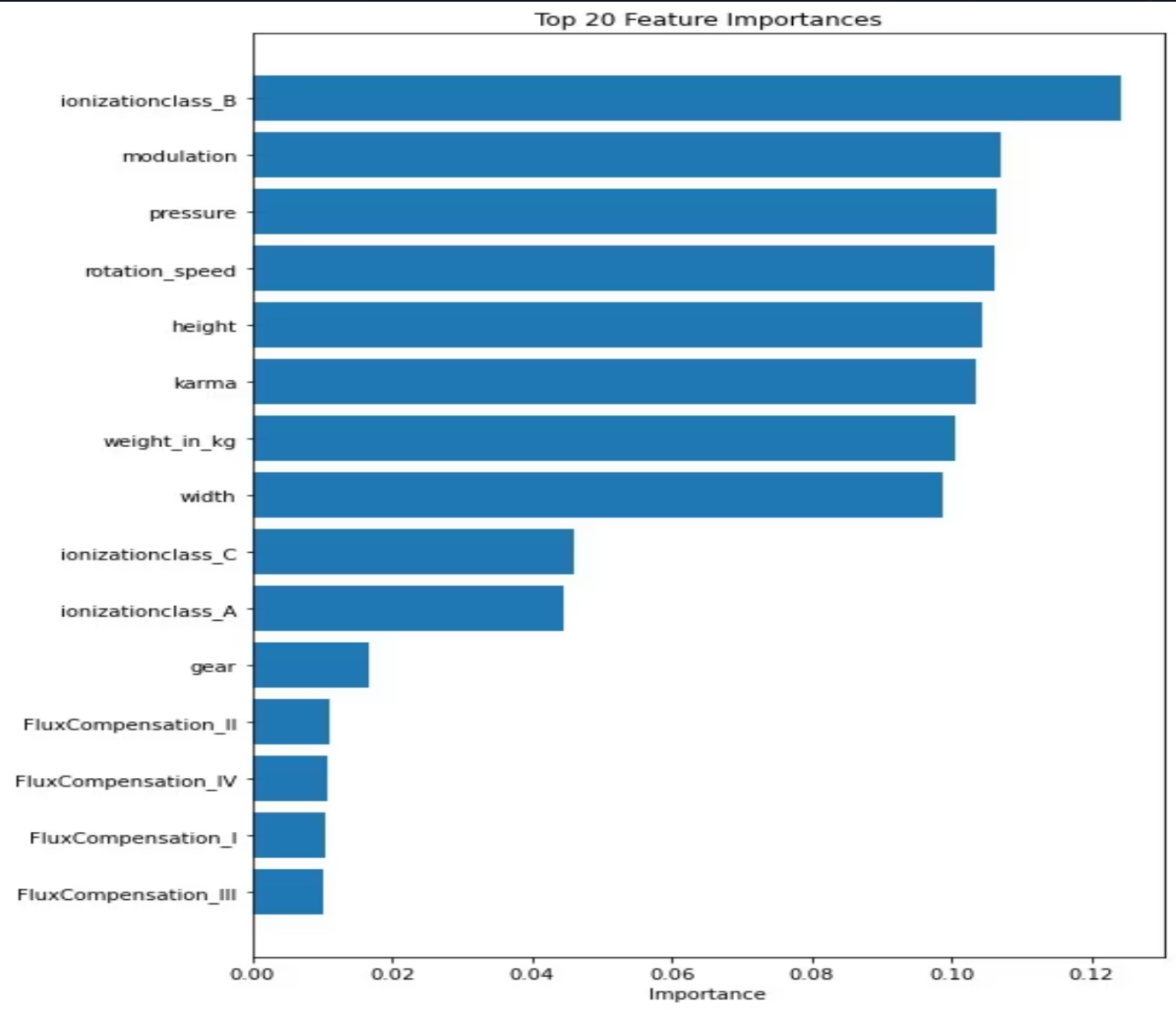
Net Savings per Batch: €126,520

By accurately predicting and preventing errors, the XGBoost model reduces costs by over 80% per manufacturing batch.

This direct financial impact underscores the immediate value of integrating predictive quality assurance.

Feature Importance: Understanding Key Drivers

Identifying the most influential factors helps SmartBuild optimize processes and allocate resources effectively.



IonizationClass_B

This sensor reading is a critical predictor of potential defects.



Modulation

Settings related to modulation also play a significant role.



Pressure

Variations in pressure heavily influence error rates.

Recommendation

Prioritize **maintenance and calibration** of Ionization sensors to proactively mitigate defect causes.

Targeting these key drivers offers the most efficient path to sustained quality improvement.

Conclusion & Next Steps: Towards a Smarter Future

01

Weight Prediction Solved

Our Polynomial Regression model delivers highly accurate weight predictions, ensuring consistent product quality.

02

Significant Defect Savings

The XGBoost model for defect prediction is projected to yield ~€126,000 in savings per batch.

03

Identified Key Drivers

We have identified critical machine settings that influence defects, enabling targeted operational improvements.

04

Pilot Integration

We recommend a pilot integration of these models into SmartBuild's existing IT infrastructure to validate real-world performance.

This data-driven approach will empower SmartBuild to achieve unprecedented levels of quality assurance and operational efficiency.

